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United States Patent	12416292
Kind Code	B2
Date of Patent	September 16, 2025
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Wind turbine offset pitch drive pinion

Abstract

A repair device is provided for a wind turbine having a bearing with damaged teeth driven by a motor. The repair device includes a carrier for connection to the bearing and radially aligned with the damaged teeth. A gear train is mounted on the carrier for creating a torque transfer path from the motor to non-damaged teeth of the bearing.

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Appl. No.:	18/524163
Filed:	November 30, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20250179995 A1	Jun. 05, 2025

Publication Classification

Int. Cl.: F03D80/50 (20160101); F03D15/10 (20160101); F03D80/70 (20160101)

U.S. Cl.:

CPC F03D80/504 (20230801); F03D15/10 (20160501); F03D80/701 (20230801);
F05B2260/40311 (20130101)

Field of Classification Search

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates generally to wind turbines, and specifically to a repair device for addressing bearing wear in a wind turbine.

BACKGROUND

(2) A wind turbine typically includes a plurality of blades connected to a nacelle through a rotor hub and a control mechanism to control the pitch of the blades. The wind turbines adjust rotor blade pitch for several reasons, namely, to improve the wind turbine efficiency by adapting the blade aerodynamics with wind speed changes. Another reason is to take the blades to an unloaded or feathered condition in the event that wind speed exceeds the maximum safe rated speed or in the event of fault conditions.

(3) To perform the pitch, each blade is provided with a bearing between the hub and the blade, and some sort of mechanism to provide the force for pitching the blade and maintaining it in a given position. This pitching arrangement enables each blade to be turned approximately 120° around its rotational axis.

SUMMARY

(4) In one example, a repair device is provided for a wind turbine having a bearing with damaged teeth driven by a motor. The repair device includes a carrier for connection to the bearing and radially aligned with the damaged teeth. A gear train is mounted on the carrier for creating a torque transfer path from the motor to non-damaged teeth of the bearing.

(5) In another example, a repair device is provided for a wind turbine having a bearing with damaged teeth driven by a motor. The repair device includes a carrier for connection to the bearing and radially aligned with the damaged teeth. A gear train is mounted on the carrier and includes a drive gear configured to receive torque from the motor. A pair of idler gears is positioned on opposite sides of the drive gear and configured to mesh with the drive gear. A pair of pinion gears is meshed with the respective idler gears and with non-damaged teeth of an inner ring of the bearing

for delivering torque from the drive gear to the non-damaged teeth of the bearing.

(6) In another example, a method is provided for installing a repair device on a wind turbine having a bearing with damaged teeth driven by a motor. The method includes providing the repair device including a gear train mounted on a carrier. The repair device is radially aligned with the damaged teeth on the bearing. The carrier is connected to the bearing such that the gear train creates a torque transfer path from the motor to non-damaged teeth of the bearing.

(7) Other objects and advantages and a fuller understanding of the invention will be had from the following detailed description and the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view of an example wind turbine in an aspect of the present invention.

(2) FIG. 2 is an enlarged view of a portion of the wind turbine of FIG. 1.

(3) FIG. 3 is a section view of FIG. 2 taken along line 3-3.

(4) FIG. 4 is a perspective and enlarged view of a bearing of the wind turbine.

(5) FIG. 5 is a side view of an alternative bearing.

(6) FIG. 6 is an exploded view of an example bearing repair device in accordance with an aspect of the invention.

(7) FIG. 7 is a schematic illustration of the bearing repair device installed on the wind turbine.

(8) FIG. 8 is a flow chart illustrating the steps of installing the repair device on a damaged bearing.

DETAILED DESCRIPTION

(9) The present invention relates generally to wind turbines, and specifically to a repair device for addressing bearing wear in a wind turbine. The wind turbine bearing to be repaired by the present invention can be, for example, a pitch bearing, a yaw bearing and/or a drive bearing. The present disclosure offers a distinct technical advantage by offering an easy, inexpensive means of re-establishing functional torque transfer between the drive motor and the toothed inner ring of the bearing when one or more teeth become damaged. To this end, a carrier is provided with a gear train including a series of gears coupled to the drive motor and arranged to circumferentially span the damaged teeth on the bearing and engage teeth that are not/less damaged, thereby maintaining functional torque transfer to the bearing.

(10) FIG. 1 illustrates an example wind turbine **100** having a bearing repair device **150** installed therein in accordance with an aspect of the present invention. The wind turbine **100** can be an offshore wind turbine or an onshore wind turbine. The wind turbine **100** includes a tower **102** extending vertically from a foundation **104** or base **104** on land or off-shore. A nacelle **106** is mounted at the upper end of the tower **102**. A rotor hub **108** is mounted at the front end of the nacelle **106** and is rotatable about an axis R.sub.1 relative to the nacelle in a generally vertical plane. Turbine blades **110** extend radially from the hub **108** and rotate with the hub in response to wind impinging upon the blades.

(11) Referring to FIG. 2, each of the blades **110** is connected to the hub **108** through a respective pitch bearing **112**. Each bearing **112** is configured to rotate its corresponding blade **110** between different pitch angles relative to an axis R.sub.2. The bearing **112** includes concentric inner and outer rings **114**, **116**. The outer ring **116** is fixed to the hub **108**. The inner ring **114** is fixed to the blade **110** and rotatably coupled to the outer ring **116**. To this end, a series of rolling elements (not shown) are provided between and interface with the rings **114**, **116** to facilitate smooth relative rotation between the rings about the axis R.sub.2. In one example, the rolling elements can be formed as spherical balls made of steel and/or other suitable materials. These rolling elements can be selectively hardened to provide long operating life and smooth rotation between inner and outer rings **114**, **116**. Lubrication can also be provided between the rings **114**, **116** to reduce friction

therebetween during operation.

(12) Turning to FIG. 3, a gear ring **118** is provided on the inner ring **114**. The gear ring **118** can be integrally formed with the inner ring **114** or secured thereto with fasteners, welding, etc. It will be appreciated that the gear ring **118** can completely encircle the rotation axis R.sub.2 (as shown) or extend less than 360° around the axis (not shown). In any case, the gear ring **118** has a radially inner side **122** and a radially outer side **124** (see also FIG. 4). Teeth **130** extend from the radially inner side **122** towards the axis R.sub.2. The radially outer side **124** abuts the inner ring **114**.

(13) A motor **140** is provided within the hub **108** and includes a shaft **142** bearing an output gear **144** rotatable with the shaft **142**. The output gear **144** is meshed with the teeth **130** on the gear ring **118** such that rotation of the motor **140** drives the output gear **144** which, in turn, rotates the inner ring **114** relative to the outer ring about the axis R.sub.2. More specifically, rotating the output gear **144** causes the gear ring **118** to move along a circumferential path P, thereby rotating the inner ring **114** secured thereto and changing the pitch of the blade **110** associated with that particular bearing **112**.

(14) One or more sensors **111** can be provided on the nacelle **106** for monitoring wind speed and/or wind direction during operation of the wind turbine **100**. A controller **115** is connected to the sensors **111** and the motor **140**. The motor **140**, in response to signals sent by the sensors **111** indicative of the wind speed and/or wind direction, can be operated by the controller **115** to adjust the pitch angle(s) of the blade(s) **110** during operation of the wind turbine **110** to help maximize its efficiency.

(15) It will be appreciated that a separate motor **140** and gear ring **118** is associated with each blade **110** such that the pitch angle of each blade can be individually adjusted. The same controller **115** and sensors **111** can be used to individually control each motor **140**. With this in mind, the circumferential or arcuate extent of any particular gear ring **118** dictates the degree to which the blade **110** associated therewith can change its pitch because the blade is coupled to the inner ring **114** on which the gear ring resides. As shown, the gear ring **118** is a ring and, thus, the blade **110** connected thereto can rotate completely about its axis R.sub.2. It will be appreciated that each gear ring **118** can extend over the same arcuate angle as any other gear ring or a different arcuate angle.

(16) It will also be appreciated that the gear ring **118** can also be provided on the radially outer side of the outer bearing **116**, as shown in FIG. 5. In this configuration, the teeth **130** extend radially outward from the outer ring **116** relative to the axis R.sub.2. The motor **140**, shaft **142**, and output gear **144** (not shown) are then positioned relative to the hub **108** to enable the output gear to mesh with the teeth **130** on the outer ring **116**. In this configuration, the blade **110** is secured to the outer ring **116** and movable therewith in response to rotation of the output gear **144**. The motor **140** can rotate the outer ring **116** relative to the inner ring **114** and about the axis R.sub.2 to adjust the pitch of the blade **110** secured to that particular outer ring. The same configuration can be utilized for the remaining bearings **112**.

(17) In either configuration, it will be appreciated that over time, one or more of the teeth **130** on the inner bearing ring **114** can become damaged, e.g., worn, chipped, cracked, corroded, etc. As a result, the interface between the motor output gear **144** and the teeth **130** can become compromised. With this in mind, the present invention provides a repair device **150** for maintaining or reestablishing a reliable, torque transferring connection between the output gear **144** and the teeth **130**.

(18) Turning to FIGS. 6-7, the repair device **150** includes a carrier or base **152** connected directly or indirectly to the bearing **112**. As shown, fasteners **166** rigidly fix the carrier **152** to the hub **108** [thereby indirectly connecting the repair device **150** to the bearing **112**] such that the repair device **150** is radially aligned with the damaged teeth (indicated as “DT”). Alternatively, the fasteners **166** can connect the carrier **152** directly to the outer ring **116** (not shown) to radially align the repair device **150** with the damaged teeth.

(19) A drive gear **151**, at least one idler gear **154**, and at least one pinion gear **156** are provided on

the carrier **152** and collectively form a gear train **158**. The gear train **158** can include multiple of each gear **154**, **156** (as shown) or one of each gear (not shown). Regardless, the number of idler gears **154** and the number of pinion gears **156** is the same. The idler gears **154** can be the same size as the pinion gears **156** or a different size, i.e., smaller or larger diameters. As shown, the drive gear **151** is the same size as the pinion gear(s) **156**. In an alternative construction, the idler gear(s) **154** are omitted and a drive chain (not shown) connects the drive gear **151** to the at least one pinion gear **156**. Consequently, the gear train in such constructions includes the drive gear **151**, the drive chain, and at least one pinion gear **156**.

(20) The drive gear **151** is intended to replace the output gear **144** of the motor **140**. Consequently, the drive gear **151** is configured to be attached to the shaft **142**, which extends through an opening **161** in the carrier **152**, and receive torque from the motor **140**. That said, the drive gear **151** can be the same size as the output gear **144** it replaces or smaller than the output gear.

(21) The idler gears **154** are rotatably supported on axles **160** extending through openings **162** in the carrier **152**. The pinion gears **156** are rotatably supported on axles **170** extending through openings **172** in the carrier **152**. As shown, the openings **162** are circumferentially aligned with one another and the openings **161**, **172** are circumferentially aligned with one another. The openings **162** are radially offset from the openings **161**, **172**. Other positions and orientations are contemplated for the openings **162** and/or the openings **161**, **172**. Regardless, the gear train **158** has a generally concave contour or centerline, i.e., a centerline extending through all the openings **161**, **172**, to accommodate the contour of the [inwardly extending] teeth **130** on the inner ring **114**.

(22) Once the output gear **144** is removed and the drive gear **151** connected to the shaft **142**, the idler gears **154** are meshed with the drive gear **151** and radially spaced from the teeth **130**. It will be appreciated that the drive gear **151** can engage the damaged teeth DT (as shown) or be spaced therefrom (not shown). Regardless, the pinion gears **156** are meshed with both the idler gears **154** as well as with the teeth **130**.

(23) As a result, torque delivered to the drive gear **151** from the motor **140** is transferred circumferentially outward to both idler gears **154**, then to both pinion gears **156**, and ultimately to the teeth **130**. This torque transfer is illustrated schematically at T.sub.1 and T.sub.2 in FIG. 7, with T.sub.1 representing clockwise torque and T.sub.2 representing counterclockwise torque when the drive gear **151** rotates in the clockwise direction. It will be appreciated that all the torque directions shown will be reversed when the drive gear **151** rotates in the counterclockwise direction (not shown). In each case, any pinion gear **156** in the gear train **158** rotates in the same direction as the drive gear **151** and any idler gear **154** rotates in the opposite direction. That said, the sizing of the gears **154**, **156** and/or positioning of the openings **162** and/or openings **172** ensures that only the gears **151**, **156** engage the teeth **130**. The idler gears **154** rotate in the opposite direction from the gears **151**, **156** and therefore are spaced from the teeth **130**.

(24) The size of each gear **151**, **154**, **156** and the size and number of each gear **154**, **156** in the gear train **158** is selected such that the arc length of the gear train **158** (indicated at s.sub.1) is greater than the arc length of the worn/damaged teeth DT (indicated at s.sub.2). In one example, the arc length s.sub.1 of the gear train **158** is less than 90°, e.g., about 30° to about 45°. The arc length s.sub.1 of the gear train **158** can be measured between the axles **170** of the outermost pinion gears **156** (as shown) or between the outermost teeth of said pinion gears (not shown). In any case, the gear train **158** is sized to circumferentially span the entirety of the damaged teeth DT on the inner ring **114** such that the pinion gears **156** (or at least one pinion gear) is always engaged with non-damaged teeth **130** as the pitch of the blade **110** is adjusted. In this manner, the repair device **150** provides a torque transfer path from the motor **140** and around the damaged teeth DT to non-damaged teeth **130**, thereby allowing the bearing **112** to maintain functionality without requiring replacement of the teeth, removal of the inner ring **114** or the so called “dentures” to be applied to the damaged teeth DT.

(25) It will be appreciated that the repair device **150** can also be adapted to transfer torque between

the motor **140** and non-damaged teeth **130** when the bearing **112** has the configuration shown in FIG. **5**. To this end, the carrier **152** and openings **161**, **162**, **172** can be arranged to orient the gear train **158** to extend generally along a convex contour or centerline to accommodate the [outwardly extending] teeth **130** on the outer ring **116**. As with the earlier described configuration, the carrier **152** used with the bearing **112** of FIG. **5** can either be directly connected to the bearing **112** (in this case directly connected to the inner ring **114**) or indirectly connected to the bearing by being secured to the hub **108**.

(26) It will also be appreciated that in either configuration of the bearing **112** the output gear **144** can be retained and the gear train **158** adapted to receive torque therefrom. In this example, the drive gear **151** is omitted and the gear train **158** is provided with only idler gears **154** and pinion gears **156** arranged on opposite sides of the output gear **144** for delivering torque from the motor **140** to non-damaged teeth **130**. The repair device **150** is then installed such that the idler gears **154** on either side of the output gear **144** are meshed therewith to allow for torque transfer from the output gear to the idler gears.

(27) Although the repair device **150** is shown in use with the pitch bearing **112** one having ordinary skill in the art will readily understand that any bearing in the wind turbine **100** can be repaired with the repair device, such as but not limited to the yaw bearing and/or drive bearing in accordance with the present invention. To this end, the repair device **150** can be modified by adjusting its size, shape, contour, number of gears, toothings, etc. to accommodate the particular bearing to be repaired.

(28) FIG. **8** is a flow chart **200** illustrating the steps of installing the repair device **150** on the bearing **112** of a wind turbine **100**. At step **202**, the repair device **150** is provided at the wind turbine **100**. This occurs whether the output gear **144** is removed or retained. At step **204**, the repair device **150** is radially aligned with the damaged teeth DT on the bearing **112**. The repair device **150** can be positioned radially within the bearing **112** (for the bearing shown in FIG. **3**) or radially outward of the bearing (for the bearing shown in FIG. **5**). At step **206**, fasteners **166** [directly or indirectly] connect the carrier **152** to the bearing **112** to fix the gear train **158** in place such that the pinions **156** engage teeth **130** on opposing circumferential sides of the damaged teeth DT to create a reliable torque transfer path from the drive gear **151**, through the idler gears **154**, through the pinion gears **156**, and to the teeth **130**. When the gear train **158** omits the drive gear **151** and the output gear **144** is retained, the drive train creates a reliable torque transfer path from the output gear **144**, through the idler gears **154**, through the pinion gears **156**, and to the teeth **130**. In both scenarios, the repair device **150** allows the motor **140** to reliably and repeatably adjust the pitch of the previously affected blade **110**.

(29) The repair device of the present invention is advantageous in that it provides an efficient, less expensive means of addressing damaged bearing teeth than current solutions. To this end, providing a repair device that can be readily adapted to the tooth issue faced in a particular bearing and that can be readily installed thereon with minimal effort alleviates the need to remove the bearing, remove the problematic teeth altogether or apply expensive and time consuming “dentures” to the damaged teeth. All the currently existing components within the hub can be maintained in their current positions while the repair device is simply secured in position in the appropriate location to enable the gear train to temporarily or permanently drive non-damaged teeth to adjust the blade pitch angle. It will be appreciated that the same repair device can be moved to another location where additional teeth on the bearing become damaged or another, different repair device can be constructed and installed.

(30) What have been described above are examples of the present invention. It is, of course, not possible to describe every conceivable combination of components or methodologies for purposes of describing the present invention, but one of ordinary skill in the art will recognize that many further combinations and permutations of the present invention are possible. Accordingly, the

present invention is intended to embrace all such alterations, modifications and variations that fall within the spirit and scope of the appended claims.

Claims

1. A repair device for a wind turbine having a hub and a bearing with damaged teeth driven by a motor, comprising: a carrier plate for connection to the bearing and radially aligned with the damaged teeth; and a gear train mounted along the length of the carrier plate for creating a torque transfer path from the motor to non-damaged teeth of the bearing, wherein the carrier plate and the gear train are separate from the bearing and the hub and attached to the hub with fasteners.
2. The repair device of claim 1, wherein the gear train receives torque from the motor and is meshed with the non-damaged teeth for delivering torque from the motor to the non-damaged teeth.
3. The repair device of claim 1, wherein the gear train comprises: a drive gear configured to receive torque from the motor; at least one idler gear configured to mesh with the drive gear; and at least one pinion gear meshed with the at least one idler gear and the non-damaged teeth of the bearing for delivering torque from the drive gear to the non-damaged teeth.
4. The repair device of claim 1, wherein the gear train extends only to one side of the drive gear.
5. The repair device of claim 1, wherein the gear train extends to opposite sides of the drive gear.
6. The repair device of claim 1, wherein the gear train comprises: a drive gear configured to receive torque from the motor; a pair of idler gears positioned on opposite sides of the drive gear and configured to mesh with the drive gear; and a pair of pinion gears meshed with the respective idler gears and with the non-damaged teeth of the bearing for delivering torque from the drive gear to the non-damaged teeth.
7. The repair device of claim 1, wherein the gear train has an arc length of less than 90°.
8. The repair device of claim 1, wherein the gear train has an arc length of about 30° to about 45°.
9. The repair device of claim 1, wherein the gear train has a concave centerline for transferring torque to the non-damaged teeth on an inner ring of the bearing.
10. The repair device of claim 1, wherein the gear train has a convex centerline for transferring torque to the non-damaged teeth on an outer ring of the bearing.
11. The repair device of claim 1, wherein the bearing is one of a pitch bearing, a drive bearing, and a yaw bearing on the wind turbine.
12. A repair device for a wind turbine having a hub and a bearing with damaged teeth driven by an output gear of a motor, comprising: a carrier plate for connection to the bearing and radially aligned with the damaged teeth; and a gear train mounted along the length of the carrier plate and extending along a concave centerline, comprising: a drive gear configured to receive torque from the motor; a pair of idler gears positioned on opposite sides of the drive gear and configured to mesh with the drive gear; and a pair of pinion gears meshed with the respective idler gears and with non-damaged teeth of an inner ring of the bearing for delivering torque from the drive gear to the non-damaged teeth of the bearing, wherein the carrier plate and the gear train are separate from the bearing and the hub and attached to the hub with fasteners.
13. The repair device of claim 12, wherein the gear train has an arc length of less than 90°.
14. The repair device of claim 12, wherein the gear train has an arc length of about 30° to about 45°.
15. A method of installing a repair device on a wind turbine having a hub and a bearing with damaged teeth driven by an output gear of a motor, the method comprising: providing the repair device comprising a gear train mounted along the length of a carrier plate separate from the bearing and the hub; removing the output gear; radially aligning the repair device with the damaged teeth on the bearing; and connecting the separate carrier plate to the hub with fasteners such that the gear train is coupled to the motor and creates a torque transfer path from the motor to non-damaged teeth of the bearing.

16. The method of claim 15, wherein the step of radially aligning the repair device comprises positioning the gear train radially inward of an inner ring of the bearing having the damaged teeth.
 17. The method of claim 15, wherein the step of radially aligning the repair device comprises positioning the gear train radially outward of an outer ring of the bearing having the damaged teeth.
 18. The method of claim 15, wherein the damaged teeth are on one of a pitch bearing, a drive bearing, and a yaw bearing on the wind turbine.
 19. The method of claim 15, wherein the carrier plate is connected to the hub after the teeth are damaged on the bearing.
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